First two pages of chapter where S1 Text excerpt was derived from (https://stat545.com/join-cheatsheet.html)

Chapter 15 Join two tables

Join (a.k.a. merge) two tables: dplyr join cheatsheet with comic characters and publishers.

15.1 Why the cheatsheet

Examples for those of us who don't speak SQL so good. There are lots of Venn diagrams re: SQL joins on the internet, but I wanted R examples. Those diagrams also utterly fail to show what's really going on vis-a-vis rows AND columns.

Other great places to read about joins:

- The dplyr vignette on Two-table verbs.
- The Relational data chapter in R for Data Science (Wickham and Grolemund 2016). Excellent diagrams.

15.2 The data

Working with two small data frames: `superheroes` and `publishers` .

```
library(tidyverse) ## dplyr provides the join functions

superheroes <- tibble::tribble(
       ~name, ~alignment,  ~gender,          ~publisher,
   "Magneto",      "bad",   "male",            "Marvel",
     "Storm",     "good", "female",            "Marvel",
  "Mystique",      "bad", "female",            "Marvel",
    "Batman",     "good",   "male",                "DC",
     "Joker",      "bad",   "male",                "DC",
  "Catwoman",      "bad", "female",                "DC",
   "Hellboy",     "good",   "male", "Dark Horse Comics"
  )

publishers <- tibble::tribble(
  ~publisher, ~yr_founded,
        "DC",       1934L,
    "Marvel",       1939L,
     "Image",       1992L
  )
```

Sorry, cheat sheet does not illustrate "multiple match" situations terribly well.

Sub-plot: watch the row and variable order of the join results for a healthy reminder of why it's dangerous to rely on any of that in an analysis.

15.3 `inner_join(superheroes, publishers)`

> `inner_join(x, y)` : Return all rows from `x` where there are matching values in `y` , and all columns from `x` and `y` . If there are multiple matches between `x` and `y` , all combination of the matches are returned. This is a mutating join.

```
(ijsp <- inner_join(superheroes, publishers))
#> Joining, by = "publisher"
#> # A tibble: 6 x 5
#>   name     alignment gender publisher yr_founded
#>   <chr>    <chr>     <chr>  <chr>          <int>
#> 1 Magneto  bad       male   Marvel          1939
#> 2 Storm    good      female Marvel          1939
#> 3 Mystique bad       female Marvel          1939
#> 4 Batman   good      male   DC              1934
#> 5 Joker    bad       male   DC              1934
#> 6 Catwoman bad       female DC              1934
```

We lose Hellboy in the join because, although he appears in `x = superheroes` , his publisher Dark Horse Comics does not appear in `y = publishers` . The join result has all variables from `x = superheroes` plus `yr_founded` , from `y` .

superheroes

name	alignment	gender	publisher
Magneto	bad	male	Marvel
Storm	good	female	Marvel
Mystique	bad	female	Marvel
Batman	good	male	DC
Joker	bad	male	DC
Catwoman	bad	female	DC
Hellboy	good	male	Dark Horse Comics

publishers

publisher	yr_founded
DC	1934
Marvel	1939
Image	1992

inner_join(x = superheroes, y = publishers)

name	alignment	gender	publisher	yr_founded
Magneto	bad	male	Marvel	1939
Storm	good	female	Marvel	1939
Mystique	bad	female	Marvel	1939
Batman	good	male	DC	1934
Joker	bad	male	DC	1934
Catwoman	bad	female	DC	1934

15.4 `semi_join(superheroes, publishers)`

> `semi_join(x, y)` : Return all rows from `x` where there are matching values in `y` , keeping just columns from `x` . A semi join differs from an inner join because an inner join will return one row of `x` for each matching row of `y` , where a semi join will never duplicate rows of `x` . This is a filtering join.